Addition and Multiplication of Matrices

Start with a potential example. This is jumping the gun a bit since we haven't defined these yet, but it's good to show from a motivational standpoint.

$$\begin{bmatrix} 1 & 2 & 3 \\ 5 & 7 & 8 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 2 \\ 4 & 6 & 8 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 5 \\ 9 & 13 & 16 \end{bmatrix}$$

So we "add" these by adding the entries in the same location and naturally maintaining that location.

This is very natural, someone who knows nothing about matrix addition would probably guess that it works that way. I.e. the two matrices must be same size and you add them component-wise. That being so, then we presumably have:

$$\begin{bmatrix} 1 & 2 & 3 \\ 5 & 7 & 8 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 & 2 \\ 4 & 6 & 8 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 6 \\ 20 & 42 & 64 \end{bmatrix}$$

Incorrect unfortunately

But why is that wrong? That, like addition, is probably the way most would guess it to work if they had no prior knowledge of it. Especially after seeing how addition works, they'd be even more inclined to think that.

But what you know of matrix multiplication even from the prior chapter, in the restricted setting where the second matrix was only permitted to have one column when we were trying to solve $Ax = b$ style systems, you know that's not how matrix multiplication works.
But why?

As alluded these work very differently from each other. Addition is again very natural in how it works, most people with no prior knowledge of how it works would likely correctly guess how it works. Multiplication is anything but natural in comparison though.

For both, to better understand how they work, think of an $m \times n$ matrix as a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Note that if we are taking two or more transformations (which we naturally are since we are talking about adding or multiplying), we are not necessarily assuming the m and n values are the same amongst all the transformations.

<u>Addition:</u> When we add two functions f and g, logically they must have overlapping domains as say $(f + g)(b)$, which equals $f(b) + g(b)$, that only makes sense if both $f(b)$ and $g(b)$ individually make sense, i.e. b is in the the domain of both.

Likewise $f(b)$ and $g(b)$ can only be added if they are in same space so f and g probably must have same codomain or close as well.

Thus to add two matrices A and B, their associated linear transformations must have same domain and codomain (no overlap otherwise) meaning A and B must be exactly same size for A + B to be defined.

Big restriction, but at least how it works is very straightforward from there. Just add the matrices "component-wise", i.e. add each entry to its corresponding entry in same location in other matrix.

So size of sum is the same as that of A and B. And very natural to add matrices that way as adding two vectors in $\mathbb{R}^m$ is done component wise, this addition of matrices will have same effect naturally so makes sense.

<u>Multiplication:</u> Don't need to say much more about addition, multiplication is much trickier and more important as well.

Many might be familiar with how it's defined, they have seen the algorithm for it. But why is it defined that way, certain rows of one times certain columns of other to put it shortly. But where is the motivation to define it like that?

Good to clarify that, many mistakes made in multiplying matrices, and understanding where it comes from likely will help you avoid mistakes.

How to define multiplication of matrices? Maybe associate it with the product of linear transformations like we associated addition of matrices with the sum of linear transformations?

Big problem with that! Consider two linear transformations from $\mathbb{R}$ to $\mathbb{R}$:

$$j(x) = 3x \quad , \quad k(x) = 2x$$

The product of these transformations would be the function $6x^2$ which is of course not a linear transformation so can't be a matrix transformation!

And that's in the simplest case when we are going from $\mathbb{R}$ to $\mathbb{R}$, if our codomain is some $\mathbb{R}^m$ how even to "multiply" two vectors there, not clear!

So how could we define the product of the two associated matrices to be another matrix given that it can't realistically be correlated with the product of the associated linear transformations (assuming that product is even defined to begin with!).

As alluded could you maybe define product of matrices only when A, B are same size and do the multiplication component-wise as in addition? After all as said, that is what many would probably guess for how matrix multiplication would work, especially seeing how sum of matrices works.

And doing that, unlike trying to think of product of matrices as product of the two associated linear transformations, would at least define make that "product" into a linear transformation. But which one? How does it relate to the two original linear transformations?

In sum the new matrix represented the sum of the two associated linear transformations. But here, that potential product won't have a meaningful association with the two original unfortunately.

So while you could define product of matrices that way, it won't be helpful from a practical standpoint so it wouldn't be useful to do so.

Hence we figure out something more practically useful in terms of product.

Something that we showed last time (1.9.44) is composition of two linear transformations is a linear transformation still if that composition is defined. So that can be expressed as a matrix, maybe can associate product of matrices with that?

Say $S: \mathbb{R}^p \dashrightarrow \mathbb{R}^n$, $T: \mathbb{R}^n \dashrightarrow \mathbb{R}^m$

As done last time, $T \circ S$ is defined here since range of S is contained in domain of T. Say A is the matrix for T and B the matrix for S. Then A is $m \times n$ and B is $n \times p$.

So if we are going to define multiplication AB in association with composition of the linear transformations, that can only be done if the number of columns of A equals the number of rows of B. Assuming that is so how to actually compute AB? Say

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

$$B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}$$

Suppose we multiply A by first column of B only. We are familiar with how to do that based on solving $Ax = b$ systems in last chapter. So first row of A "times" that column (as noted earlier, that's called dot product).

Then second row of A times that column, continuing creates an $m \times 1$ matrix just as when solving $Ax = b$ as b was an $m \times 1$ matrix in that context too. So make that $m \times 1$ column the first column of AB.

Then multiply A by second column of B and make that the second column of AB. And so on like that will create the full AB. This will be $m \times p$ since each column we created was $m \times 1$, and collectively there will be p columns since that's how many B had.

But why define this way? Many might think when multiplying two matrices that we do it component-wise (showed earlier why that might not be desirable), others might think do rows times rows instead of rows times columns, former seems more symmetrical. The association of this with composition of linear transformations in following example makes this clearer.

Say

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad , \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Given the sizes we could do rows of A times rows of B if we so desired. But let's see how it works in the linear transformations. What is A (Bx), again that's composition.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \left(\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5x_1 + 6x_2 \\ 7x_1 + 8x_2 \end{bmatrix}$$

$$= \begin{bmatrix} 5x_1 + 6x_2 + 2(7x_1 + 8x_2) \\ 3(7x_1 + 8x_2) + 4(7x_1 + 8x_2) \end{bmatrix}$$

$$= \begin{bmatrix} 19x_1 + 22x_2 \\ 43x_1 + 50x_2 \end{bmatrix}$$

Now suppose we tried to do this via multiplying the matrices A and B first, i.e. computing the composition function then applying it to the input.

$$(AB)x = \left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \right) \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} 19x_1 + 22x_2 \\ 43x_1 + 50x_2 \end{bmatrix}$$

Same thing as we got in other method. So that's why multiplying matrices in the way we do meshes with this. If we did AB via row times row or multiplying the the individual components similar as we do in addition it definitely would not work!

This also inherently tells us another very important property of matrix multiplication, that it is <u>associative</u>. That is, for all A, B, C where this makes sense,

$$(AB)C = A(BC)$$

which makes sense since composition of functions is known to be associative.

However, we also have that matrix multiplication is not <u>commutative</u>, i.e.

$$AB \neq BA$$

at least not always. Also makes sense since composition of functions is not necessarily commutative in fact sometimes only one of AB, BA is even defined. But even if both are defined might not be the same size, for example:

$$A = \begin{bmatrix} 0 & 1 \\ 2 & 3 \\ 4 & 5 \end{bmatrix} \quad , \quad B = \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \end{bmatrix}$$

Then

$$AB = \begin{bmatrix} 3 & 4 & 5 \\ 9 & 14 & 19 \\ 1 & 24 & 29 \end{bmatrix} \quad , \quad BA = \begin{bmatrix} 10 & 13 \\ 28 & 40 \end{bmatrix}$$

Both defined but not same size. Only way both defined and same size is both square of same size. But even then not necessarily equal, for example

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad , \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Then

$$AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad , \quad BA = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Even though same size, still not equal to each other.

Not only does it emphasize that fact, it also shows a product of two nonzero matrices can be zero matrix.

Plus that we can have one of AB, BA as zero while the other is not. Definitely not like real number multiplication in either respect!

Say one asks "If $xy = 0$, does that force $x = 0$ or $y = 0$? Many would probably say yes, and indeed in the context of x, y being real numbers that would be true. But if x, y are matrices then this is not necessarily true, again context is very important!

Even though matrix multiplication is quite different from real number multiplication, it still does carry over some important properties like it's relationship to addition, namely distributive law, i.e.

$$A(B + C) = AB + AC$$

In general when dealing with a new "operation", like matrix multiplication, whether that operation is commutative and/or associative is quite relevant.

This is common in higher level math, and most often think commutativity is more important, but actually associativity is much more important. Why?

Well say we have some operation *, then what is

$$a * b * c * d * e$$

Thing is if * is not associative, this is totally unclear. Would need to add many parentheses to make it clear what order we are performing the operations in which makes it very annoying. In short, an operation that is not associative is very difficult to work with which is why associativity is so important to have in an operation.

But why does matrix multiplication logically correlate with composition of linear transformations? Even though matrix multiplication is very different than real number multiplication, it still inherently involves that as part of the process. For instance

$$f(x) = x^2 \quad , \quad g(x) = x + 1$$

The product of those is nothing like their composition (whichever order we do composition in) so again why does that somehow correlate with matrices?

Big key: that only works with linear transformations! The f and g given are definitely are not such!

Say we take instead

$$j(x) = 3x \quad , \quad k(x) = 2x$$

Then

$$(j \circ k)(x) = j(k(x)) = j(2x) = 3(2x) = 6x$$

So for these linear transformations, the correlation with product, i.e. product of matrices $[3]$ and $[2]$ is clear. So in that restricted type of setting such correlation does exist.

For any matrix A, a very important related matrix is its <u>transpose</u>. This is denoted by A^T

<u>Transpose</u>: Matrix which switches rows and columns of the original matrix. I.e. first row becomes first column and vice versa. Likewise for second row and second column and beyond.

An $m \times n$ matrix has transpose which is $n \times m$.

Early Transpose Properties:

(1) $(A + B)^T = A^T + B^T$

This is quite apparent. More relevant is:

(2) $(AB)^T = B^T A^T$

Well known but why is this true? Prove via composition of functions maybe? Let's show it via straight matrix multiplication ideas. Show that each entry in one matrix matches its corresponding one in the other.

For any i, j show that the ij-th entry matches, that again being the entry in i-th row, j-th column.

In $(AB)^T$ this entry is same as ji-th entry in AB by definition of transpose. And via matrix multiplication, that is dot product of j-th row in A with i-th column in B.

Meanwhile in $B^T A^T$ the ij-th entry is the dot product of the i-th row in B^T with j-th column in A^T by how matrix multiplication is defined. Then by definition of transpose, that is dot product of i-th column in B with j-th row in A.

Thus in both, that entry is created same way so those do match hence that being an arbitrary entry, those matrices match completely proving they are equal.

Transpose is used in many settings, mostly in the later, harder, linear algebra applications. For now just generic calculation ideas with it really.

But a few ideas. One we are often interested in if the columns of A are linearly independent which is determined by if there is a nonpivot column. Or if the columns of A span the appropriate $\mathbb{R}^m$ which comes down to whether every row of A has a pivot.

So if we are investigating these properties in A could we check instead A^T reversing what they stand for since transpose switches rows and columns?

Perhaps so but why do that? Well to check pivots we need REF. If A is not in REF or even close, say A is lower triangular, it might be annoying to convert it. But in that context A^T is upper triangular so likely in REF or very close. So that might make it easier.

Transpose also switches domain and codomain of A when thinking of each matrix as a linear transformation.

In that regard might it be similar to the "inverse" function of the linear transformation that A represents should such an inverse exist? Not equal to it but maybe perhaps some correlation?

Remember inverse only exists if function is one-to-one. So assuming linear transformation is such, what about finding its inverse? That's the next section!